

Does implied volatility efficiently predict future realized volatility?

*And how does the answer differ between
Bitcoin and the S&P 500?*

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Does implied volatility predict future realized volatility?

Same answer for Bitcoin and the S&P 500?

- Implied volatility is the market's forecast of future risk.
- In equities, the relationship has been studied for thirty years.
- In crypto, it has barely been documented.

What we know vs. what we don't

Equity markets

- Implied volatility is informative
Christensen & Prabhala (1998)
- Slope below one, positive premium
Bollerslev, Tauchen & Zhou (2009)
- Predictive power weakens in crises
Whaley (2009), Giot (2005)

Crypto markets

- DVOL exists only since March 2021
- No published predictive evidence
- No regime-conditional analysis of
BTC option markets

Our contribution: apply the equity benchmark to crypto, in a unified framework, with regime splits.

Five hypotheses

- **H1.** Implied volatility overestimates future realized volatility on average: positive variance risk premium.
- **H2.** Implied volatility is informative but not a perfect predictor.
- **H3.** The variance risk premium is larger in crypto than in equities.
- **H4.** The predictive relationship weakens in stress periods.
- **H5.** Equity markets are close to efficient in normal periods; crypto is not.

Data

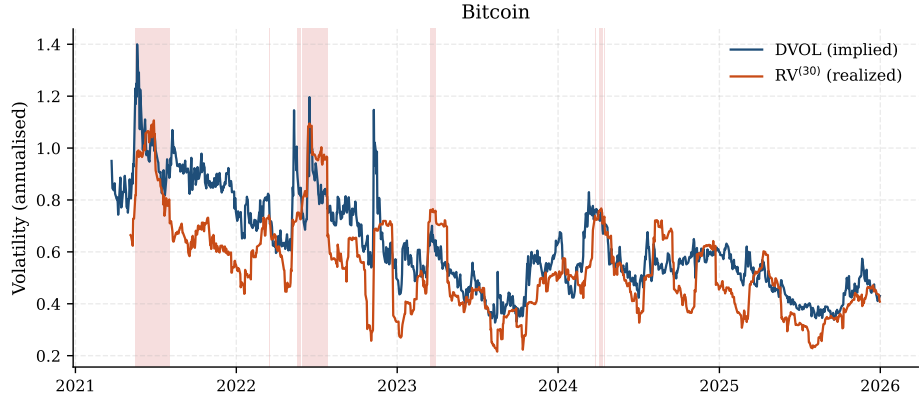
Series	Provider	Identifier
Bitcoin spot	Binance public API	BTCUSDT daily
Bitcoin implied vol.	Deribit public API	BTC_DVOL
S&P 500 close	Yahoo Finance (yfinance)	GSPC
VIX close	Yahoo Finance (yfinance)	VIX

- **Period:** 24 March 2021 → 31 December 2025.
- **After NYSE-calendar inner join:** 1 201 daily observations.
- All series come from public APIs (no proprietary terminal data).

Method, in one slide

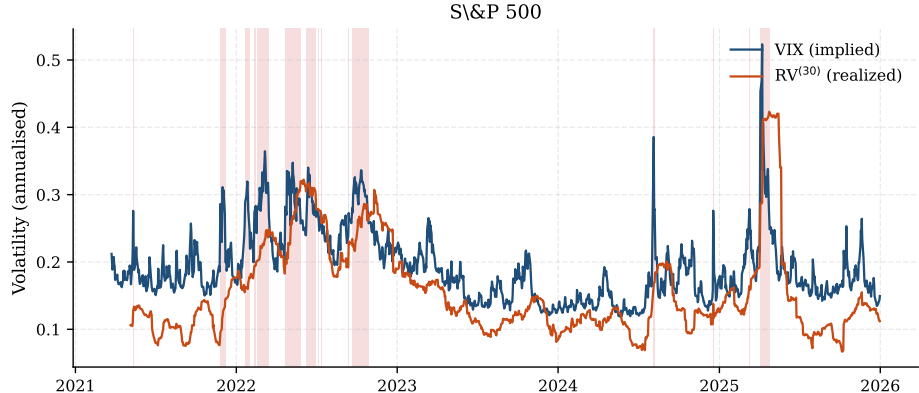
- **Realized volatility:** annualised standard deviation of 30-day log returns.
- **Predictive regression:** future realized volatility regressed on current implied volatility.
Slope = 1 means efficient.
- **Regimes:** *stress* = top 10 % of VIX (equities) or BTC realized volatility (crypto), set ex ante.
- **Inference:** Newey–West HAC standard errors to correct for rolling-window overlap.

A first look: Bitcoin



- DVOL and 30-day realized volatility move together, with a positive average wedge.
- Red shading: stress days (BTC 30-day RV above its 90th percentile, set ex ante).

A first look: S&P 500



- Volatility runs in a much narrower band than Bitcoin, but spikes sharply in crises.
- Red shading: stress days (VIX above its 90th percentile, set ex ante).

Implied volatility is informative, but biased

	Bitcoin	S&P 500
Slope $\hat{\beta}$	0.55***	0.79***
HAC s.e.	(0.09)	(0.10)
R^2	0.33	0.36
$H_0 : \beta = 1$	rejected ($p < 0.001$)	rejected ($p = 0.03$)
N	1 171	1 171

- Both slopes are highly significant: implied volatility carries real information.
- Both are significantly below one: positive variance risk premium in both markets.

*** significant at 1 %. Standard errors: Newey–West HAC, 30 lags.

The variance risk premium is larger in crypto



Crypto investors require more than twice the compensation per unit of volatility risk.

Confirms H3.

Headline result

In calm markets, the VIX looks almost efficient.

$$\hat{\beta} = 0.97$$

S&P 500, normal regime, $N = 955$

$$p = 0.89 \text{ for } H_0 : \beta = 1$$

We cannot reject that the VIX is a fully efficient predictor of S&P 500 realized volatility in normal periods.

The premium reacts to stress in opposite directions

	Normal	Stress	Direction
Bitcoin	+0.087	+0.055	<i>compresses</i>
S&P 500	+0.032	+0.047	<i>widens</i>

- **Equity stress** $\Rightarrow$ premium *widens*. Implied volatility surges with hedging demand.
- **Crypto stress** $\Rightarrow$ premium *compresses*. Realized volatility catches up faster than DVOL.

An asymmetry only visible when both markets are studied together.

Why? Three economic mechanisms

- 1 **Hedging demand.** Equity investors buy out-of-the-money puts in stress, which pushes the VIX above expected realized volatility.
- 2 **24/7 trading.** Bitcoin realized volatility adjusts to news within hours, not days, which compresses the wedge faster.
- 3 **Market maturity.** Equity option markets are deep and dominated by institutions; crypto options are concentrated on a few venues and a more retail base.

What we don't claim

- **Short sample.** DVOL only goes back to March 2021.
- **Aggregated indices.** No full option surface, so no skew and no term structure.
- **Daily frequency.** Intraday data would refine realized volatility.
- **Asymmetric regime definition.** VIX for equity stress vs. realized volatility for crypto stress.

Take-aways

- 1 Implied volatility is a **risk-adjusted expectation**, not a raw forecast, in both markets.
- 2 Equity markets are **close to efficient in normal periods**; crypto markets are **not**.
- 3 The variance risk premium **widens in equity stress** and **compresses in crypto stress**. This asymmetry is only visible across markets.

Code, data and full pipeline:
github.com/X9ClementX9/iv-rv-thesis

Thank you.

Questions?



Replication materials

github.com/X9ClementX9/iv-rv-thesis

Thesis, source code, dataset and HAC estimation routines